Modulation of FUS RRM Folding by the Neighboring Disordered Region

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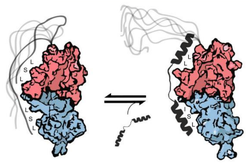
Background and Motivation

Intrinsically Disordered Regions (IDRs) next to Folded Domains

IDRs are **highly prevalent** in eukaryotic proteomes, interacting with broad spectrum of biomolecular partners

Musselman, Catherine A., and Tatiana G. Kutateladze. *Iscience* 24.2 (2021).



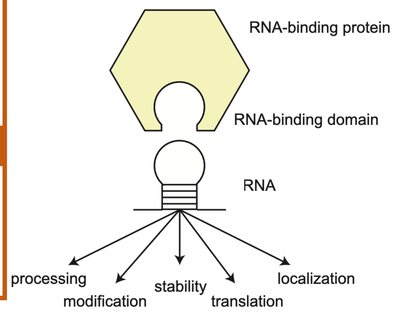
About 90% of human proteome IDRs are **adjacent to folded domains**, exerting a diverse range of effects.

Staby, Lasse, et al. *Journal of Molecular Biology* 433.24 (2021): 167320.

Van Der Lee, Robin, et al. Chemical reviews 114.13 (2014): 6589-6631.

Fused in sarcoma (FUS) as an example of RNA-Binding Proteins

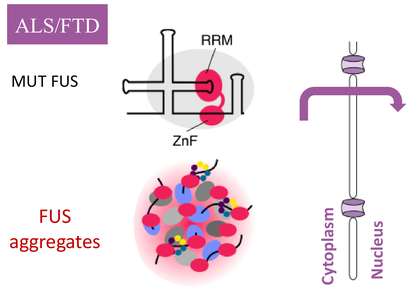
FUS plays a key role in **RNA splicing**, especially through interactions with U1 small nuclear (sn) RNA.



Hentze, Matthias W., et al. *Nature reviews Molecular cell biology* 19.5 (2018): 327-341.

FUS aggregation is a pathological hallmark of Amyotrophic Lateral Sclerosis (ALS) and Frontotemporal Dementia (FTD)

ALS/FTD



>> Impaired snRNP assembly

>> Downstream defects in RNA processing

>> Motor neuron degeneration

Jutzi, Daniel, et al. *Nature Communications* 11.1 (2020): 6341.

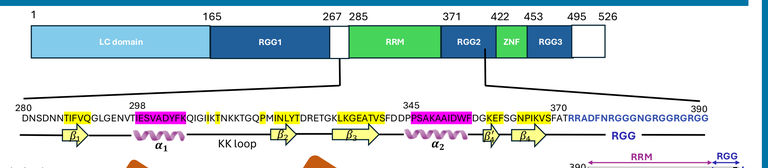
The **folding stability** of the FUS RNA-recognition motif (RRM) is biophysically critical, as its **irreversible unfolding** triggers aggregation.

The RRM domain is significantly perturbed by the presence of its neighboring domains.

What regions specifically are involved in inter-domain interactions of FUS?

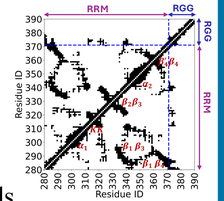
Lu, Yimei, Liangzhong Lim, and Jianxing Song. *Scientific reports* 7.1 (2017): 1043.

Simulation Methods



>> Initial structures:

Taken from PDB ID 6SNJ



Cutoff 8 A° for contact maps

>> CHARMM36 and TIP3P modified water force fields

>> Sampling method:

>>> MM-OPES (Multithermal-Multiumbrella On-the-fly Probability Enhanced Sampling)

>> Order parameter (Q: fraction of native-like contacts)

>> 150 mM NaCl

>> 500 ns considered as equilibration

Q Based on the distance between each atomic pair considered in contact

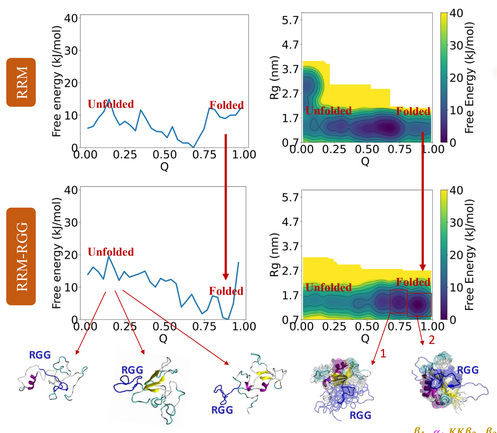
$$Q = \frac{1}{N_{\mathrm{nb}_{\mathrm{stem}}}} \sum_{(i,j)} \frac{1}{1 + \exp(\gamma(r_{ij} - \lambda r_{ij}^{0}))}$$

Q =1

- Totally **Folded** RRM

Results and Discussions

Structural changes of RRM in the presence of RGG



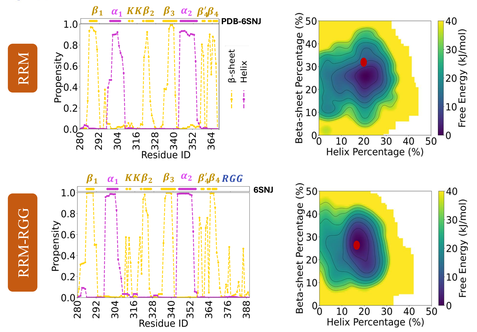
Folded RRM is **more stable** in the presence of RGG

Same compactness between folded state of RRM and RRM-RGG

>> Projecting onto secondary structure fractions

Lower fractions of helix, β-sheet, and KK loop formation in the isolated RRM

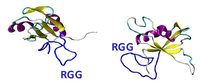
How does RGG mediate folding?

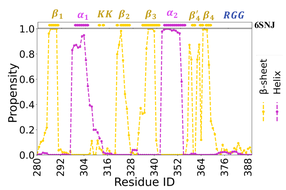


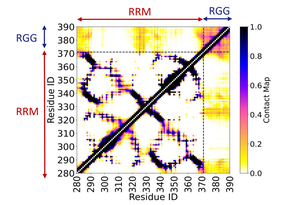
How does RGG help secondary structure formation?

>> Analyzing two folded basins in RRM-RGG

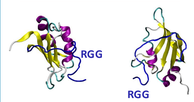
The most populated cluster in basin 1 (Q ≈ 0.76)

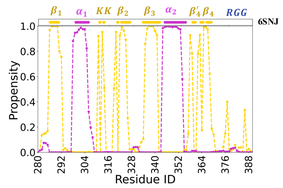


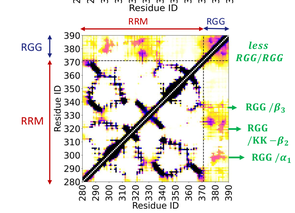




The most populated cluster in basin 2 (Q ≈ 0.90)







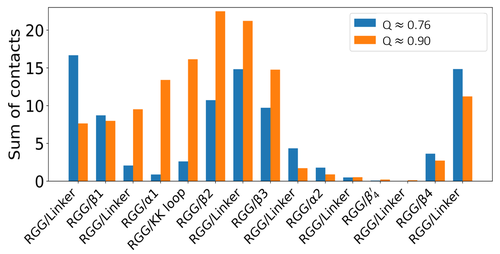
RGG/RRM contacts stabilize RRM secondary structures.

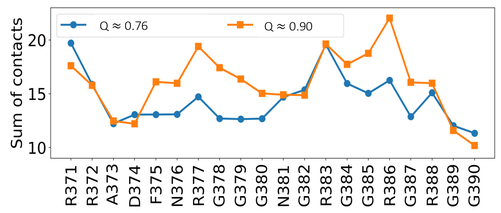
RGG acts as a **stabilizing belt** for RRM in Q ≈ 0.90

>> Taking sum over different regional interactions and for each RGG residue

The increase of contacts in areas from the linker of $\beta_1 - \alpha_1$ to β_3 in Q ≈ 0.90 compared to Q ≈ 0.76

Largest increases in contact frequency for residues spanning F375–N381, and G384–G387





Conclusions

>> The interactions between a disordered region and an adjacent globular domain increases the folding stability for FUS.

>> In the FUS protein, the RRM achieves the maximum folding stability (Q > 0.9) in the presence of RGG region.

>> The presence of the RGG segment still yield compact configurations even in the unfolded state.

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